

Dynamic Graph Neural Policies for Multi-Robot Cooperative Navigation: Edge Features, Safety Filtering, and Density Limits

Xuelin Hu

Liuzhou Railway Vocational Technical College
Liuzhou 545000, China
huxuelinai@gmail.com; huxl@ltzy.edu.cn

Youjing Fu

College of Earth and Environmental Sciences
Lanzhou University
Lanzhou 730000, China
fuyj2025@lzu.edu.cn

Xiaoqin Fu*

Liuzhou Railway Vocational Technical College
Liuzhou 545000, China
xiaoqin.fu@qq.com; fuxiaoqin@ltzy.edu.cn

Jingchao Wang

School of Computer Science
Zhongyuan University of Technology
Zhengzhou 450007, China
static@zut.edu.cn

Abstract—Multi-robot navigation in shared space requires simultaneous goal reaching, coordination, and collision avoidance under a communication topology that changes at every control step. We study whether a shared message-passing policy over a dynamic communication graph can deliver such behavior, and we report where it fails. Robots are modelled as graph nodes, edges are established by a communication radius, and each directed edge carries a five-dimensional relative-motion feature that encodes normalized relative position, relative velocity, and closing speed. Policies are trained by imitation from an ORCA-style expert with a joint objective combining action, one-step separation, goal-progress, and smoothness terms, and are optionally followed by an external pairwise safety filter. We evaluate A* grid tracking, ORCA-style and RVO-style local controllers, an independent per-robot MLP, a message-passing GNN, a neighbor-attention GNN, and a graph transformer over four map families, 5–40 robots, and 30 unseen test seeds, with ROS 2, Gazebo, and RViz2 integration verified on a Wheeltec platform. Message passing improves task completion over an edge-free policy of identical capacity (0.725 versus 0.425 success at 10 robots), but the mean-aggregation GNN does not surpass the independent MLP in the main comparison, and adding neighbor attention lowers success to 0.342. Multi-scale training with edge features raises goal success to 0.958–0.967 at 5–10 robots under safety filtering, yet strict no-physical-collision success falls from 0.708 at 20 robots to 0.250 at 30 and 0.000 at 40. We therefore identify 10–20 robots as the practical operating range of the current system and treat higher densities as a stress test that exposes a safety bottleneck which imitation learning alone does not remove.

Index Terms—multi-robot systems, graph neural networks, cooperative navigation, collision avoidance, dynamic interaction graphs, ROS 2.

This work was supported by the Science and Technology Research Project of Henan Province (grant no. 262102211055) and the Key Scientific Research Project of Colleges and Universities in Henan Province (grant no. 27AQ520018).

*Corresponding author: Xiaoqin Fu.

I. INTRODUCTION

Cooperative navigation asks a group of robots to reach assigned goals in a shared workspace without colliding with each other or with static obstacles. Two properties make this harder than single-robot planning: the joint state space grows with the population, and the influence each robot has on the others is time-varying, since neighbors continuously enter and leave sensing range. Centralized joint search stays exact but scales poorly, while fully independent local policies discard the interaction structure that makes the problem tractable at all.

Graph neural networks (GNNs) offer a middle ground. A robot group can be represented as a graph whose nodes are robots and whose edges are communication relations, and message passing then aggregates neighbor information while sharing parameters across nodes. Because the same weights apply at every node, one policy can serve a variable number of robots; because edges are rebuilt from current positions, the representation adapts to a changing topology.

This paper asks two questions. Given identical observations, action limits, and termination rules, does an explicit local communication graph improve a learned policy relative to one that sees only its own node features? And how do communication range and an external safety filter trade off among goal completion, collision risk, and per-step computation?

Our contributions are:

- a reproducible framework unifying scenario generation, classical and learned baselines, training, evaluation, and visualization under one data contract;
- a message-passing policy with five-dimensional relative-motion edge features, a joint imitation and separation objective, and an external pairwise safety filter ablated separately from the policy;

- a comparison spanning A*, ORCA-style, RVO-style, an independent MLP, a mean-aggregation GNN, a neighbor-attention GNN, and a graph transformer over four map families, 5–40 robots, and 30 unseen seeds, with edge, radius, and safety ablations;
- a ROS 2, Gazebo, and RViz2 integration on a Wheeltec platform validating namespaced odometry, LiDAR, velocity-command, and marker topics.

We do not frame the results as a general superiority claim for GNNs. Several measurements contradict that reading, and Section VI reports them as measured.

II. RELATED WORK

A. Classical and Rule-Based Planning

A* searches a discretized grid heuristically and is complete on a static map, but coordinating one path per robot reintroduces combinatorial coupling [1]. Velocity-obstacle methods instead reason in velocity space, rejecting a candidate velocity if the relative motion it induces would intersect a neighbor’s trajectory [2]. They are cheap and geometrically interpretable, but depend strongly on the time horizon and on the reciprocity assumed between agents.

B. Learned Multi-Robot Collision Avoidance

Decentralized deep reinforcement learning removes joint search by acting from local observations, though dense interactions outside the training distribution remain a failure mode [3], [4]. Curriculum learning grows the difficulty of constrained passages [5], physics-informed priors encourage safe behavior [6], and semi-supervised variants cut the labeling requirement [7]. Vessel and formation applications show that communication and shape constraints fold into the same learning problem [8], [9].

C. Graph Representations for Robot Teams

Diao *et al.* encode spatial relations in a graph for path planning [10], and He *et al.* pair graph attention with hierarchical motion planning to select the neighbors that matter [11]. Goarin and Loianno use a GNN for decentralized goal assignment, which suits variable team sizes but does not address continuous avoidance [12], and Yang *et al.* couple a GNN with PPO for multi-UAV planning [13]. Surveys note the recurring tension that rule-based methods are interpretable but brittle while learned methods are flexible but lack hard guarantees [14].

Our design follows that tension’s suggested compromise: a graph policy learns cooperative motion and a separate filter handles close-range closure risk. We ablate the two independently and report strict physical-collision metrics for each rather than attributing filtered behavior to the network. We also note that reported gains are frequently established at a single population size on a few maps; the multi-scale stress test below probes that gap.

III. PROBLEM FORMULATION

Let $\mathcal{V} = \{1, \dots, N\}$ index the robots, with position $p_i \in \mathbb{R}^2$, velocity $v_i \in \mathbb{R}^2$, and goal $g_i \in \mathbb{R}^2$. A directed communication graph is rebuilt at every step from a radius R_c :

$$A_{ij} = \begin{cases} 1, & i \neq j, \|p_i - p_j\| \leq R_c, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Node features are eight-dimensional: goal-relative position, normalized velocity, direction and distance to the nearest obstacle, and a bias term. The policy emits a two-dimensional local velocity u_i with $\|u_i\| \leq v_{\max} = 1.4$ m/s, and positions evolve as $p_i^{t+1} = p_i^t + \Delta t u_i^t$ with $\Delta t = 0.08$ s.

Two distance thresholds are used throughout and never conflated. A *safety violation* occurs when any pair is closer than $d_s = 0.72$ m; a *physical collision* occurs when any pair is closer than the sum of the robot radii, 0.56 m. A *collision event* is counted each time a pair transitions from a non-violating to a violating state, so a sustained overlap contributes one event rather than one per step. Task success requires the final mean goal error across all robots to fall below a fixed threshold.

To keep quantities of different scale comparable inside the network, define the normalized operator $\bar{x} = x/s_x$. For a directed edge (j, i) we use the five-dimensional feature

$$e_{ij}^t = \left[\frac{p_i^t - p_j^t}{[W, H]}, \frac{v_i^t - v_j^t}{v_{\max}}, \frac{-((v_i^t - v_j^t) \cdot \hat{r}_{ij}^t)}{v_{\max}} \right], \quad (2)$$

with $\hat{r}_{ij}^t = (p_i^t - p_j^t)/(\|p_i^t - p_j^t\| + \epsilon)$ and $[W, H]$ the map extents. The final component is the closing speed, positive precisely when the pair is approaching, so the network can condition on motion tendency rather than instantaneous distance alone.

IV. METHOD

A. System Overview

Fig. 1 shows the closed loop. Each robot is a node, edges follow the communication radius, and the encoder, message-passing layers, and action head share parameters across nodes, so one policy handles any population size without retraining. Observations are assembled from the current state, the graph is rebuilt, and the safety filter optionally corrects the emitted velocity before execution.

B. Baselines

We implement A* grid tracking, an ORCA-style local repulsion controller, and a dependency-free RVO-style controller that enumerates candidate velocities over a finite lattice of headings and speed levels, scoring each by predicted neighbor clearance, obstacle occupancy, and goal progress. Because it exists to give every method a common substrate, we call it RVO-style and claim no equivalence to a formally verified RVO reference library. The independent MLP applies the same perceptron to every node and never receives the adjacency matrix, isolating the contribution of explicit interaction modelling from that of the node features themselves.

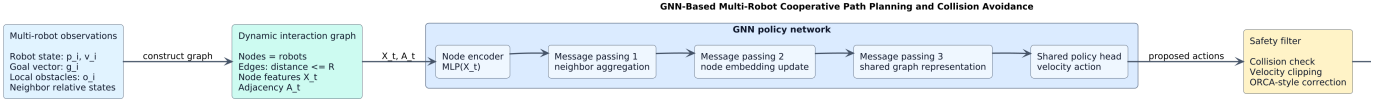


Fig. 1. Dynamic-graph message-passing policy. Node and five-dimensional edge features enter shared message-passing layers; the shared action head emits a bounded local velocity that the optional safety filter may correct.

C. Message-Passing Policy

A linear layer first maps node features to a hidden representation. Each layer averages neighbor representations and updates the node from its own state concatenated with the transformed neighborhood message:

$$m_i^{(l)} = \frac{1}{\max(1, d_i)} \sum_j A_{ij} h_j^{(l)}, \quad h_i^{(l+1)} = \sigma(W_l[h_i^{(l)} \parallel \phi_l(m_i^{(l)})]). \quad (3)$$

With edge features the message becomes

$$m_i^{(l)} = \frac{1}{\max(1, d_i)} \sum_j A_{ij} (\phi_l(h_j^{(l)}) + \psi_l(e_{ij})), \quad (4)$$

followed by $h_i^{(l+1)} = \text{ReLU}(W_l[h_i^{(l)} \parallel m_i^{(l)}])$. We use three message-passing layers, 128 hidden units, and a shared action head, with the action bounded by $\hat{u}_i = v_{\max} \tanh(f_\theta(h_i))$.

D. Attention Variants

To test whether neighbor weighting or global self-attention changes the picture, we add two models with identical inputs, outputs, data, and test scenarios. The neighbor-attention GNN replaces mean aggregation with multi-head attention over queries, keys, and values, still masked by the local adjacency. The graph transformer uses multi-head self-attention with the dynamic adjacency as mask, residual connections, and feed-forward blocks. Both use 128 hidden units, four heads, and a shared policy head.

E. Training Objective

Pure action imitation reproduces the expert’s local behavior but does not guarantee a safe closed loop, so we train on a weighted sum of four terms:

$$\mathcal{L} = \lambda_a \mathcal{L}_{act} + \lambda_c \mathcal{L}_{col} + \lambda_p \mathcal{L}_{prog} + \lambda_s \mathcal{L}_{smooth}, \quad (5)$$

where $\mathcal{L}_{act} = \frac{1}{|V|} \sum_i \text{Huber}(u_i, u_i^*)$ is the imitation term. Writing the one-step predicted pair distance as $d'_{ij} = \|p_i + \Delta t u_i - p_j - \Delta t u_j\|$, $\mathcal{L}_{col} = \frac{1}{N(N-1)} \sum_{i \neq j} [d_s - d'_{ij}]_+^2$ penalizes predicted separation. The remaining terms are

$$\begin{aligned} \mathcal{L}_{prog} &= \frac{1}{N} \sum_i [\|g_i - p'_i\| - \|g_i - p_i\|]_+, \\ \mathcal{L}_{smooth} &= \frac{1}{N} \sum_i \|u_i^t - u_i^{t-1}\|_2^2, \end{aligned} \quad (6)$$

with $[x]_+ = \max(x, 0)$. Padding nodes are masked by $q_i \in \{0, 1\}$, so every node loss becomes $\sum_i q_i \ell_i / \max(1, \sum_i q_i)$ and a graph padded to 40 nodes trains on exactly the real robots in it.

Multi-scale training samples the 5–40 robot datasets from all four map families and pads each graph to 40 nodes. The progress and smoothness terms matter in practice: a separation penalty alone reduces goal progress, so its weight must be scheduled or paired with $\mathcal{L}_{prog}$.

F. Safety Filter

After the policy emits its action, the filter inspects every close pair and applies a symmetric correction before clipping. Let $n_{ij} = \delta_{ij} / (\|\delta_{ij}\| + \epsilon)$ with $\delta_{ij} = p_i - p_j$. If $\|\delta_{ij}\| < d_s + \tau$ and $(u_i - u_j) \cdot n_{ij} < 0$, then

$$\begin{aligned} \tilde{u}_i &= u_i + \frac{\kappa_{ij}}{2} n_{ij}, \quad \tilde{u}_j = u_j - \frac{\kappa_{ij}}{2} n_{ij}, \\ \kappa_{ij} &= -(u_i - u_j) \cdot n_{ij} + [d_s + \tau - \|\delta_{ij}\|]_+, \end{aligned} \quad (7)$$

after which each $\tilde{u}_i$ is clipped to $v_{\max}$. The module runs only at inference and in the ROS control path, and Section VI reports its effect separately from the policy’s.

V. EXPERIMENTAL SETUP

Experiments span four map families—*crossing*, *open*, *random*, and *narrow*—and 5, 10, 20, 30, and 40 robots. Development data uses fixed seeds with 20 episodes of at most 160 steps each. The frozen main evaluation covers $4 \times 3 \times 5 \times 10 = 600$ episodes over three robot counts, five methods, and ten fixed test seeds; the extended learned-model comparison covers 480 episodes for four learned policies; and the multi-scale test covers 1800 safety-filtered and 1800 unfiltered episodes over 30 unseen seeds (100–129). All methods share initial states, goals, obstacles, action limits, and termination rules. Data are stored as compressed NPZ archives holding features, adjacency, actions, positions, rewards, obstacles, layout, expert identifier, and step index.

Reported metrics are goal success rate, strict no-physical-collision success rate, safety-violation events, physical collision events, minimum inter-robot separation, path length, final goal error, and mean per-step control time. Means, standard deviations, and 95% confidence intervals are computed over episodes, and violations at 0.72 m and physical collisions at 0.56 m are always reported in separate columns. Training used one 24 GB GPU with batch size 128.

VI. RESULTS

A. Main Comparison

Table I aggregates the frozen 600-episode comparison. The classical controllers are strong: A* reaches every goal, and RVO-style reaches 0.975–1.000 success with the shortest paths, but its control time grows almost two orders of magnitude across the range, from 7.8 ms at 5 robots to 87.4 ms at 20. ORCA-style keeps the largest minimum separation (0.396 m at 5 robots) at the lowest per-step cost, though some narrow passages stall it. The independent MLP is the strongest learned policy here at 0.725–0.875 success.

The mean-aggregation GNN does not beat that MLP. It reaches 0.625, 0.725, and 0.700 success at 5, 10, and 20 robots

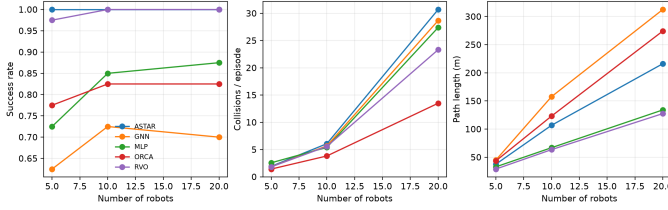


Fig. 2. Aggregate comparison across methods and robot counts on the frozen 600-episode evaluation. Classical controllers dominate success and path length; learned policies trade completion for lower per-step cost.

with comparable collision-event counts, and its path lengths and control times are the highest among learned methods. This checkpoint was trained on 10-robot data only, so the 5 and 20-robot columns are cross-scale generalization rather than matched training conditions. The conclusion we draw is narrow but firm: on this benchmark, adding a communication graph is not by itself sufficient to outperform an independent policy, and success rate or path length alone would not justify a superiority claim.

B. Attention and Transformer Variants

Table II compares four learned policies on 480 episodes. Overall goal success is 0.817 for the MLP, 0.683 for the GNN, 0.558 for the graph transformer, and 0.342 for the neighbor-attention GNN; collision-free success under the 0.56 m threshold is 0.017, 0.008, 0.058, and 0.025. Two observations follow. The transformer roughly doubles the GNN’s strict success while costing about $1.6\times$ the control time, so expressiveness helps even while the safety outcome stays poor. But adding attention in its neighbor-masked form hurts badly, cutting success to 0.342 and raising control time to 1.298 ms. Attention is not a free improvement, and no variant here is safe on its own.

C. Graph Structure Ablation

Removing edges isolates the contribution of message passing (Fig. 3a). At 10 robots the local-graph policy reaches 0.725 success against 0.425 for the edge-free policy of identical capacity; at 20 robots the figures are 0.700 and 0.425, and at 5 robots 0.625 against 0.500. The global graph, connecting every pair regardless of distance, does not improve on the local graph and is worse at 10 robots (0.625 versus 0.725). Message passing therefore contributes real coordination ability, but a denser graph is not automatically better and neither variant removes collisions.

D. Safety Filter Ablation

The safety filter is evaluated on its own against the raw policy on 60 episodes (Fig. 3b). At 10 robots it reduces mean collision events from 4.900 to 0.200 and raises mean minimum separation from 0.069 m to 0.794 m, but success drops from 1.000 to 0.900. At 20 robots collision events fall from 27.500 to 6.900 with success falling from 0.900 to 0.700, and separation reaches only 0.067 m, so no stable collision-free arrival is achieved at that density. The filter also

leaves some robots stalled rather than rerouting them. The trade is direct: separation improves substantially while task completion degrades, which is our main reason for arguing that the safety constraint belongs in training rather than in a post-hoc projection.

E. Communication Radius

Larger radii increase mean graph degree and supply more neighbors at the cost of more aggregation work. At 10 robots a radius of 2.0 m yields only 0.200 success with mean degree 0.463, while 3.4 m and 5.5 m both reach 1.000 success with degrees 2.283 and 4.420; the global graph reaches 0.900 with degree 9.000. At 5 robots success climbs monotonically from 0.100 at 2.0 m to 1.000 for the global graph, indicating that sparse graphs simply disconnect small teams. Control time stays near 1.0 ms across radii, so the connectivity gain is nearly free below 20 robots; the global graph does not, however, eliminate physical collisions.

F. Multi-Scale Stress Test

Table III reports the 1800-episode safety-filtered test over 30 unseen seeds, and it is the most informative result in this paper. Goal success is high and stable at 5–20 robots: 0.967, 0.958, and 0.758 for the GNN and 0.983, 1.000, and 0.842 for the transformer. Strict collision-free success, however, falls from 0.967 at 5 robots to 0.708 at 20, 0.250 at 30, and 0.000 at 40 for the GNN, with the transformer showing the same shape (0.983, 1.000, 0.767, 0.242, 0.017). Physical collision events per episode rise from zero at 5–10 robots to 6.883 for the GNN and 9.025 for the transformer at 40 robots, and minimum separation collapses from above 0.95 m to about 0.4 m. Control time grows steadily with population, from 0.890 ms to 2.066 ms for the GNN.

Without the filter the picture is starker: the GNN reaches 0.992–1.000 goal success at every scale but strict collision-free success of 0.092 at 5 robots and 0.000 everywhere else, and the transformer never exceeds 0.117. High arrival rates therefore coexist with near-total failure of the safety criterion, which is exactly why we report the two separately. We read 10–20 robots as the practical operating range of the current system and treat 30–40 robots as a stress test: adding robots barely degrades goal seeking, but it overloads the mechanism meant to keep them apart.

G. Seed Stability

Three independently trained GNN checkpoints give mean success of 0.517 ± 0.388 , 0.767 ± 0.161 , and 0.850 ± 0.115 at 5, 10, and 20 robots. The 5-robot variance is large enough that reporting a single best checkpoint would be misleading and indicates the present training configuration is not yet stable at small team sizes. All tables above aggregate over seeds and checkpoints rather than selecting the best run.

VII. ROS 2 INTEGRATION

A Tkinter two-dimensional interface switches between the ORCA, RVO, A*, MLP, and GNN controllers and displays

TABLE I

MAIN COMPARISON OVER FOUR MAP FAMILIES AND TEN FIXED TEST SEEDS (40 EPISODES PER CELL). SUCC: GOAL SUCCESS. COLL: COLLISION EVENTS, COUNTING TRANSITIONS INTO A < 0.72 M SAFETY VIOLATION. SEP: MINIMUM INTER-ROBOT SEPARATION (M). T : MEAN PER-STEP CONTROL TIME (MS).

Method	5 robots				10 robots				20 robots			
	Succ	Coll	Sep	T	Succ	Coll	Sep	T	Succ	Coll	Sep	T
A*	1.000	1.925	0.280	0.512	1.000	6.075	0.098	1.205	1.000	30.700	0.035	2.493
ORCA-style	0.775	1.425	0.396	0.213	0.825	3.825	0.296	0.486	0.825	13.500	0.230	1.303
RVO-style	0.975	1.825	0.294	7.779	1.000	5.575	0.131	24.941	1.000	23.325	0.062	87.373
MLP	0.725	2.575	0.191	0.342	0.850	5.400	0.095	0.483	0.875	27.400	0.038	0.775
GNN	0.625	1.850	0.224	0.683	0.725	5.675	0.149	0.828	0.700	28.675	0.041	1.129

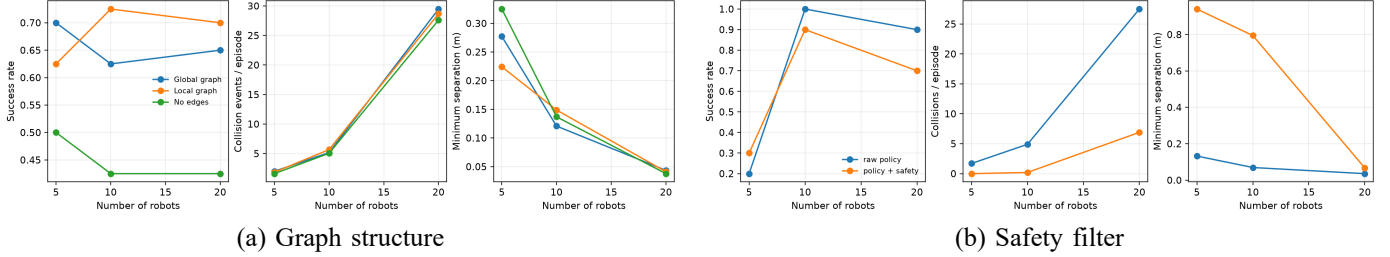


Fig. 3. Ablations. (a) Edge-free, local-radius, and global connectivity at 5, 10, and 20 robots: message passing contributes real coordination ability while a denser graph is not automatically better. (b) Raw against filtered policy, showing the separation gain versus the success loss.

TABLE II

LEARNED-POLICY COMPARISON (480 EPISODES). STRICT SUCCESS REQUIRES GOAL ARRIVAL WITH NO PAIR EVER CLOSER THAN THE 0.56 M PHYSICAL DIAMETER.

Model	Success	Strict succ.	Phys. coll. events	Min. sep. (m)	Time (ms)
MLP	0.817	0.017	9.250	0.108	0.515
GNN	0.683	0.008	8.825	0.138	0.856
Neighbor-attention	0.342	0.025	8.167	0.259	1.298
Graph transformer	0.558	0.058	8.833	0.221	1.365

TABLE III

MULTI-SCALE TEST WITH THE SAFETY FILTER OVER 30 UNSEEN SEEDS (1800 EPISODES), AGGREGATED OVER FOUR MAP FAMILIES AND INDEXED BY ROBOT COUNT. STRICT SUCCESS REQUIRES GOAL ARRIVAL WITH NO PHYSICAL COLLISION.

Model	5	10	20	30	40
<i>Goal success / strict collision-free success</i>					
GNN	0.967/0.967	0.958/0.958	0.758/0.708	0.400/0.250	0.042/0.000
Transformer	0.983/0.983	1.000/1.000	0.842/0.767	0.467/0.242	0.108/0.017
<i>Physical collision events per episode</i>					
GNN	0.000	0.000	0.075	1.158	6.883
Transformer	0.000	0.000	0.217	2.267	9.025
<i>Minimum separation (m) / control time (ms)</i>					
GNN	0.97/0.89	0.95/1.04	0.87/1.37	0.61/1.70	0.41/2.07
Transformer	1.00/1.50	0.94/1.65	0.80/1.99	0.56/2.33	0.34/2.69

robots, goals, communication edges, trajectories, and live metrics. The ROS 2 adapter targets a Wheeltec model and was verified across three namespaces on the `/robot_i/odom`, `/robot_i/scan`, and `/robot_i/cmd_vel` topics, with a marker node publishing `/multi_robot/markers` so RViz2 renders robots, goals, trails, and graph edges; the launch script is `ros2/run_multi_robot_gnn.sh`. This validates the topic contract and visualization path on a Gazebo model, not physical-robot behavior, and no real-hardware safety claim is made here.

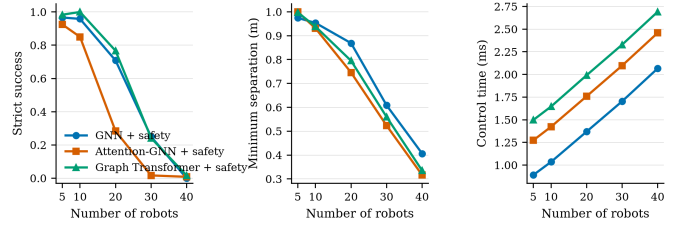
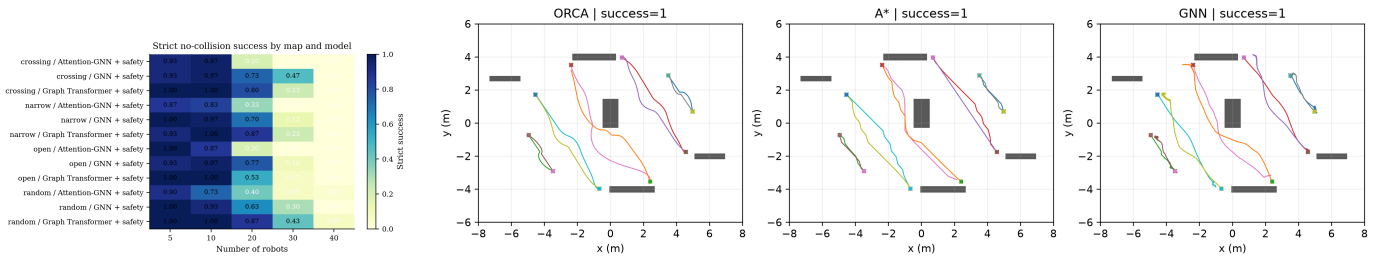


Fig. 4. Strict collision-free success, minimum separation, and per-step control time against population size under safety filtering. Goal success stays high while the safety criterion collapses above 20 robots.

VIII. LIMITATIONS AND CONCLUSION

We built and evaluated a dynamic-graph message-passing framework for multi-robot cooperative navigation, combining five-dimensional relative-motion edge features, a joint imitation and separation objective, an external pairwise safety filter, and a verified ROS 2 path. The measurements support three conclusions. Message passing contributes genuine coordination ability: the same architecture with edges removed loses roughly 0.30 success at 10 robots. A denser graph is not automatically better, since the global graph fails to improve on the local one. Most importantly, filtering and imitation together deliver high arrival rates at 5–20 robots but cannot hold the safety criterion at 30–40 robots, where strict collision-free success collapses to near zero.

Several caveats bound these conclusions. Our RVO-style controller is a finite-lattice method written for experimental uniformity, not a formally verified implementation of the RVO reference algorithm, so comparisons against it should be read accordingly. The main-comparison GNN checkpoint was trained at 10 robots, making its 5 and 20-robot numbers cross-scale generalization results rather than matched performance.



(a) Strict success by map and density

(b) Representative trajectories

Fig. 5. (a) Strict collision-free success by map family and population size; *random* and *narrow* layouts are the most demanding at high density. (b) Representative trajectories, with communication edges drawn as transient links between robots within the radius.

The imitated expert is itself rule-based, so the policy inherits its teacher's limitations, and the Gazebo work demonstrates the interface and visualization chain only rather than physical-robot behavior. Finally, the present work is imitation-based; reinforcement fine-tuning with long-horizon progress and hard safety constraints remains incomplete, and the multi-scale results are the evidence that it is needed. The clear next steps are therefore safety-constrained policy training instead of post-hoc projection, a verified RVO baseline, improved cross-map generalization, and validation on physical robots.

REFERENCES

- [1] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
- [2] P. Fiorini and Z. Shiller, "Motion planning in dynamic environments using velocity obstacles," *The International Journal of Robotics Research*, vol. 17, no. 7, pp. 760–772, 1998.
- [3] P. Long, T. Fanl, X. Liao, W. Liu, H. Zhang, and J. Pan, "Towards optimally decentralized multi-robot collision avoidance via deep reinforcement learning," *IEEE*, 2018, pp. 6252–6259.
- [4] N. Thumiger and M. Deghat, "A multi-agent deep reinforcement learning approach for practical decentralized uav collision avoidance," *IEEE Control Systems Letters*, vol. 6, pp. 2174–2179, 2022.
- [5] M. M. R. Komol, B. Tidd, W. Browne, F. Maire, J. Williams, and D. Howard, "Learning behaviours for decentralised multi-robot collision avoidance in constrained pathways using curriculum reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8538–8545, 2025.
- [6] P. Feng, R. Shi, S. Wang, J. Liang, X. Yu, S. Li, and W. Wu, "Safe and efficient multi-agent collision avoidance with physics-informed reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11 138–11 145, 2024.
- [7] M. Parooei, M. T. Masouleh, and A. Kalhor, "Semi-supervised deep reinforcement learning in decentralized multi-agent collision avoidance and path planning in a complex environment," *IEEE*, 2023.
- [8] Y. Niu, F. Zhu, M. Wei, Y. Du, and P. Zhai, "A multi-ship collision avoidance algorithm using data-driven multi-agent deep reinforcement learning," *Journal of Marine Science and Engineering*, vol. 11, no. 11, p. 2101, 2023.
- [9] Y. Wang and Y. Zhao, "Multiple ships cooperative navigation and collision avoidance using multi-agent reinforcement learning with communication," *Ocean Engineering*, vol. 320, p. 120244, 2025.
- [10] X. Diao, W. Chi, and J. Wang, "Graph neural network based method for robot path planning," *Biomimetic Intelligence and Robotics*, vol. 4, no. 1, p. 100147, 2024.
- [11] X. He, X. Shi, J. Hu, and Y. Wang, "Multi-robot navigation with graph attention neural network and hierarchical motion planning," *Journal of Intelligent & Robotic Systems*, vol. 109, no. 2, 2023.
- [12] M. Goarin and G. Loianno, "Graph neural network for decentralized multi-robot goal assignment," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4051–4058, 2024.
- [13] Q. Yang, J. Lu, Y. Zhang, and S. Shao, "Multi-uav cooperative path planning via graph neural network and proximal policy optimization," *IEEE Access*, vol. 13, pp. 193 575–193 588, 2025.
- [14] G. Yang, L. An, and C. Zhao, "Collision/obstacle avoidance coordination of multi-robot systems: A survey," *Actuators*, vol. 14, no. 2, p. 85, 2025.